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Design and Development of an Android Application for Habit Improvement and Personal Tracking: Alter

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1. Introduction

The rapid evolution of mobile technologies has led to the widespread use of Android applications, offering innovative solutions across various sectors. This project focuses on designing and developing an Android app that addresses a specific need—such as helping users better track and manage their habits.
The report begins with an overview of the project and its features, followed by a detailed look at the app's design and implementation, highlighting key technological choices.
In today's digital age, mobile devices like smartphones and tablets have become essential for communication, entertainment, and productivity. However, this constant connectivity also contributes to behavioral addictions such as excessive screen time, gaming, or social media use, and even substance dependence. These issues, increasingly common across all age groups, can lead to anxiety, depression, and social isolation.
As technology continues to evolve, it's crucial to promote healthier digital habits, raise awareness about these risks, and provide tools for users to regain balance in their lives.

2. Project Presentation

Our application, **alter**, positions itself as the first social network entirely dedicated to supporting individuals in their journey to overcome addictions and bad habits. While many existing applications stop at basic tracking or static informational content, **alter** offers a comprehensive and interactive approach, placing community support at the heart of habit change.

2.2. Context and Problem Statement

In today's fast-paced society, individuals often face stress, emotional suppression, and disorganization—factors that contribute to the rise of harmful habits such as excessive screen time, smoking, or alcohol use. At the same time, maintaining good habits can be just as challenging, with many people struggling to stay consistent on their own.
To address these issues, **alter** offers an intuitive mobile solution that combines habit tracking with social engagement, encouraging users to build healthier routines while feeling supported and motivated.

2.3. Main Features

Alter is more than a habit tracker—it is a supportive community platform. Its main features include:

- **Daily Tracking Journal**: Enables users to record their progress, reflect on challenges, and identify relapses to strengthen consistency.
- **Community Spaces**: Includes public and habit-specific chat groups where users share experiences, give advice, and encourage one another.
- **Instant Messaging**: Offers private messaging for direct, personal interaction between users.

- **Smart Reminders**: Sends regular notifications to help users stay focused on their goals and avoid falling back into harmful routines.

2.4. Project Goals

The key goals of the **alter** project are to:

- Offer a user-friendly and engaging platform for habit monitoring and personal growth.
- Create a strong sense of community to reduce isolation and foster mutual support.
- Motivate users through shared journeys, discussions, and interactive features.
- Empower individuals to make lasting changes in their behaviors and lifestyle.

By blending habit tracking with social connectivity, **alter** provides a holistic tool to support long-term recovery and positive transformation.

2.5. Report Structure

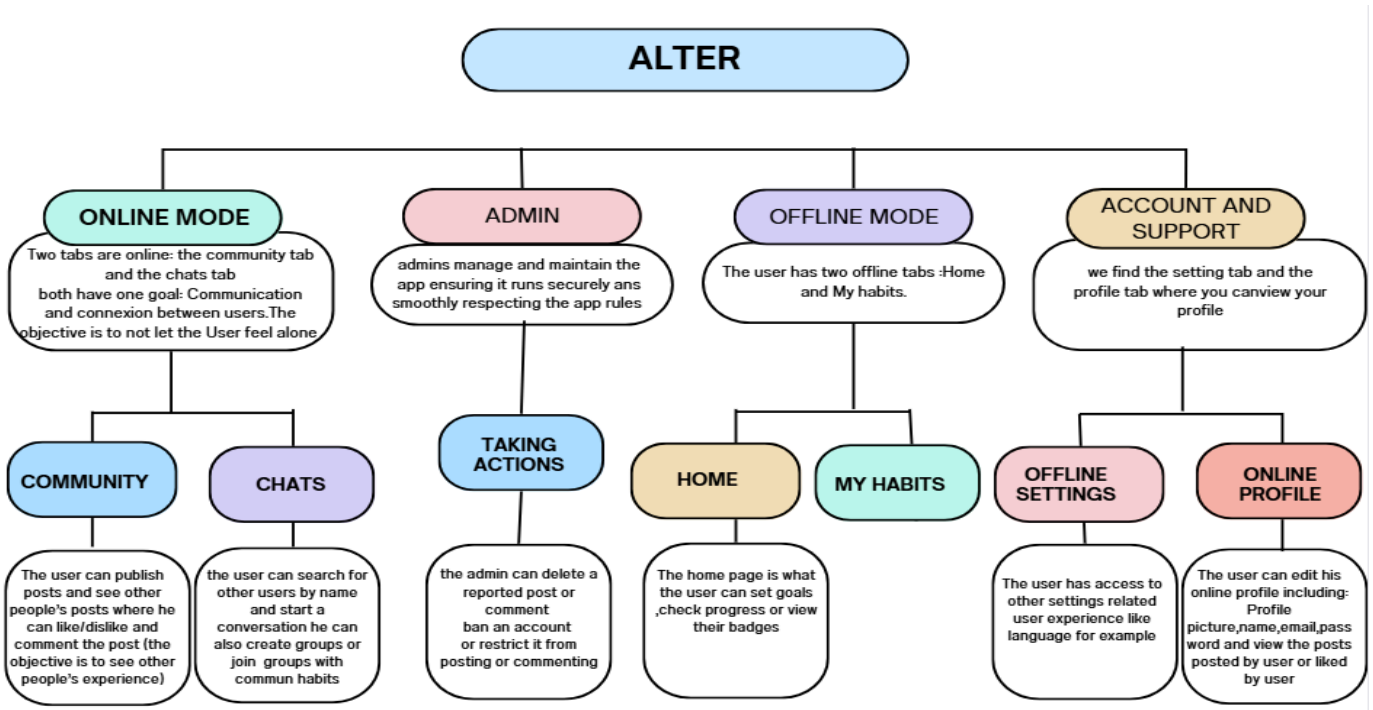
This report begins with a general presentation of the project and its core functionalities. It then details the design and development phases of the application, highlighting the technological choices and strategies that shaped the final product.

3. Conception

3.1. General Architecture

The application is composed of several key modules, each responsible for a specific aspect of functionality, such as user authentication, habit tracking, community interaction, and notifications. Communication between modules is handled efficiently to ensure seamless user experience and performance.

Below is a schematic representation of the general architecture of the application:



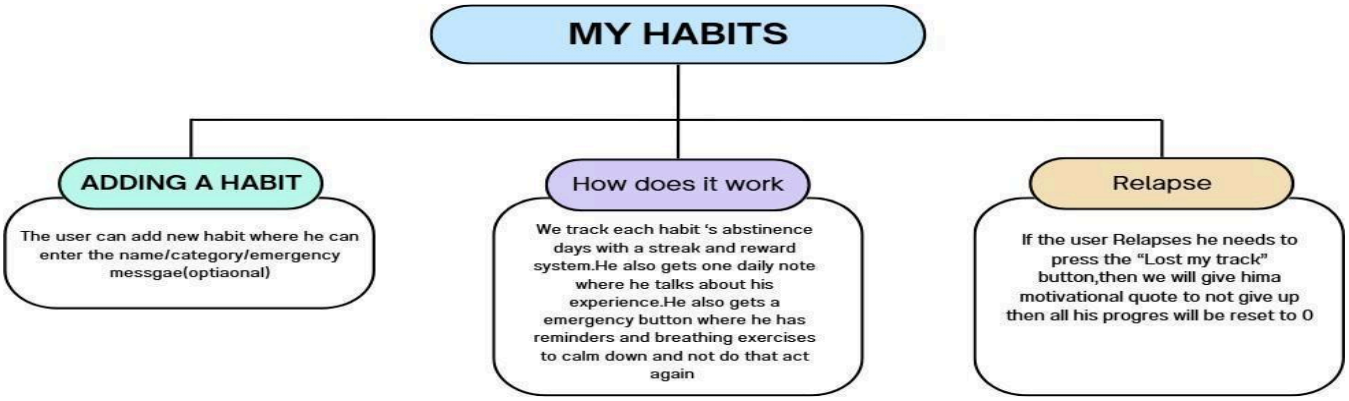


Figure 1: General architecture of **«alter»** application.

3.2. Technologies Used

Les choix technologiques sont les suivants:

- **Programming language:** Java.
- **Development environment:** Android Studio.
- **DataBase:** Firebase (onlinepart), SQLITE (offlinepart).

3.3. Modeling

The application's structure and functionality are modeled using several standard UML diagrams, which provide a clear view of the system's components and interactions.

a. *Class Diagram*

The class diagram defines the main entities of the application and their relationships, including:

- **User**: with attributes like username, email, bio, profileImage, etc.
- **Habit**: associated with the user; includes fields such as title, reminderTime, emergencyMessage, logs.
- **Log**: linked to a specific habit and date, containing mood, notes, status.
- **Post** and **Comment**: tied to the community interaction system.
- **Message**: includes sender, receiver, timestamp, and content.
- **Admin**: with extended privileges for moderation.

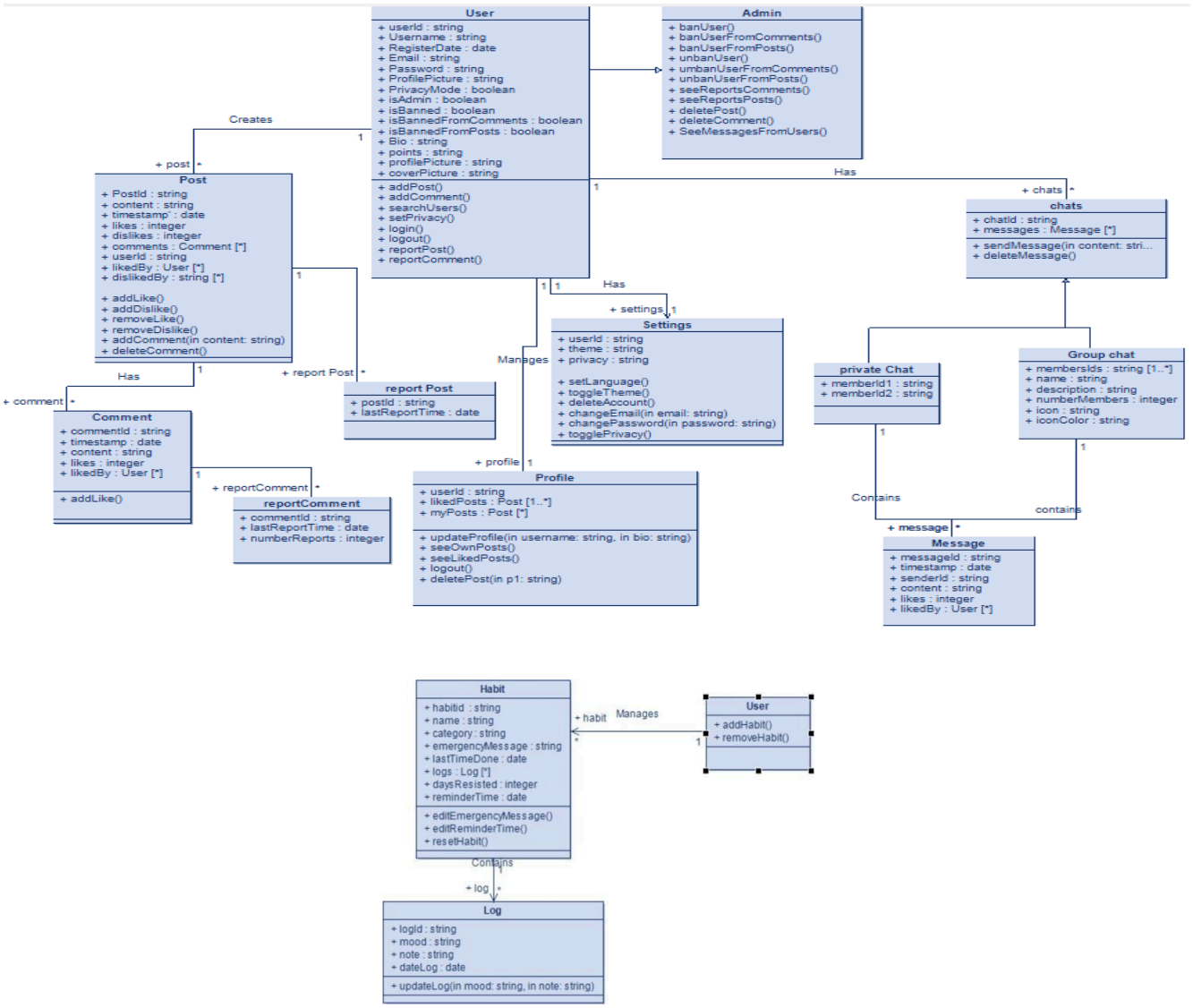


Figure 2: Class Diagram of **«alter»** application.

This diagram illustrates the object-oriented structure of the application and supports its modularity and extensibility.

b. Use Case Diagrams

The application is divided into two main functional domains:

1. Habit Tracker

Main actors: User

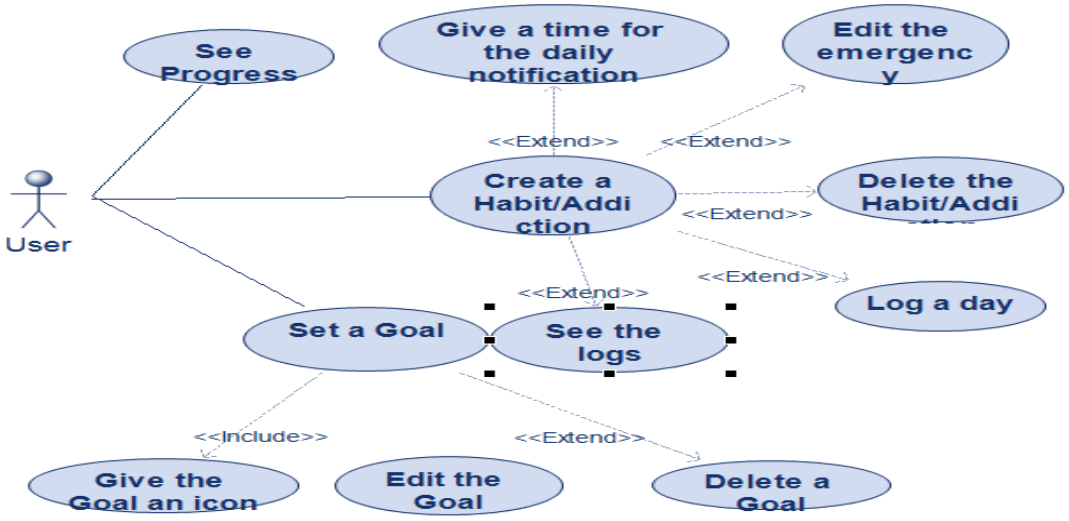


Figure 3: Habit Tracker Use cases Diagram.

Use Cases:

- addHabit(): The user creates a new habit to track.
- removeHabit(): The user removes a habit they no longer wish to follow.
- editEmergencyMessage(): The user edits the emergency message linked to a habit.
- editReminderTime(): Change the time of the habit reminder.
- resetHabit(): Reset progress counters for a specific habit.
- updateLog(): Update the habit log with mood and personal notes.

2. Community and Social Features

Actors: User, Admin

Use Cases (User side):

- addPost(): Create a new post in the community.
- addComment(): Comment on an existing post.
- searchUsers(): Search for users by name or profile.
- setPrivacy(): Adjust privacy settings.
- login()/logout(): Authenticate and disconnect from the app.
- reportPost()/reportComment(): Report inappropriate content.
- updateProfile(): Modify profile information (username, bio, image).
- seeOwnPosts()/seeLikedPosts(): View your own posts or those you've liked.
- sendMessage(): Send a private or group message.
- deleteMessage(): Delete a message from a chat.

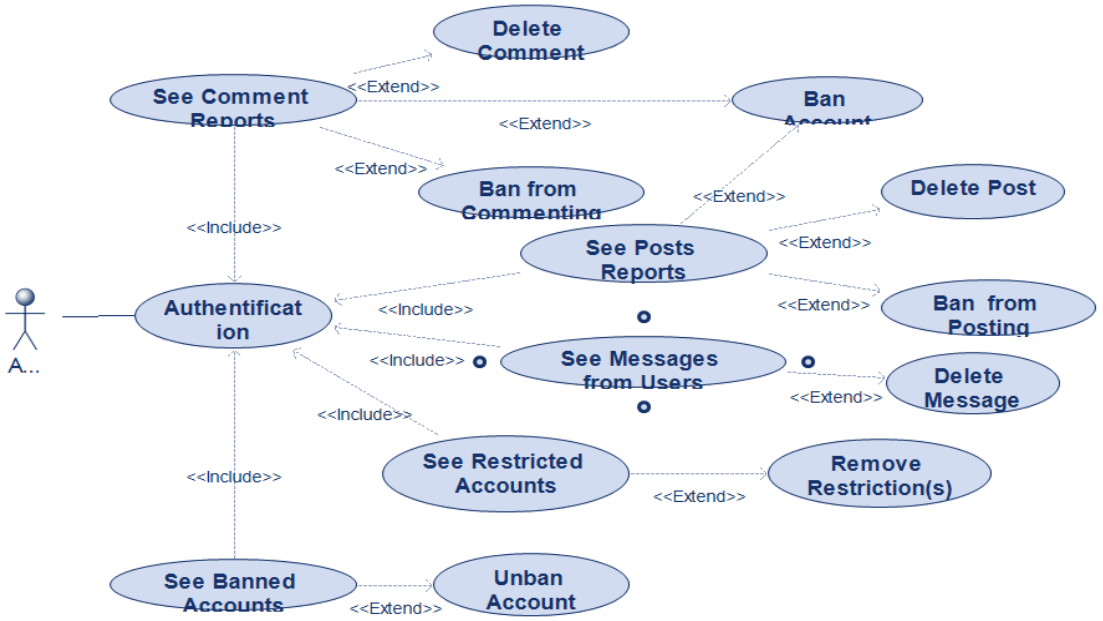


Figure 4: Community and Social Features Use cases Diagram, User side.

Use Cases (Admin side):

- banUser()/unbanUser(): Temporarily or permanently block a user.
- banUserFromPosts()/FromComments(): Restrict user actions related to posts/comments.
- seeReportedPosts()/Comments(): View reported content.
- deletePost()/deleteComment(): Remove inappropriate content.

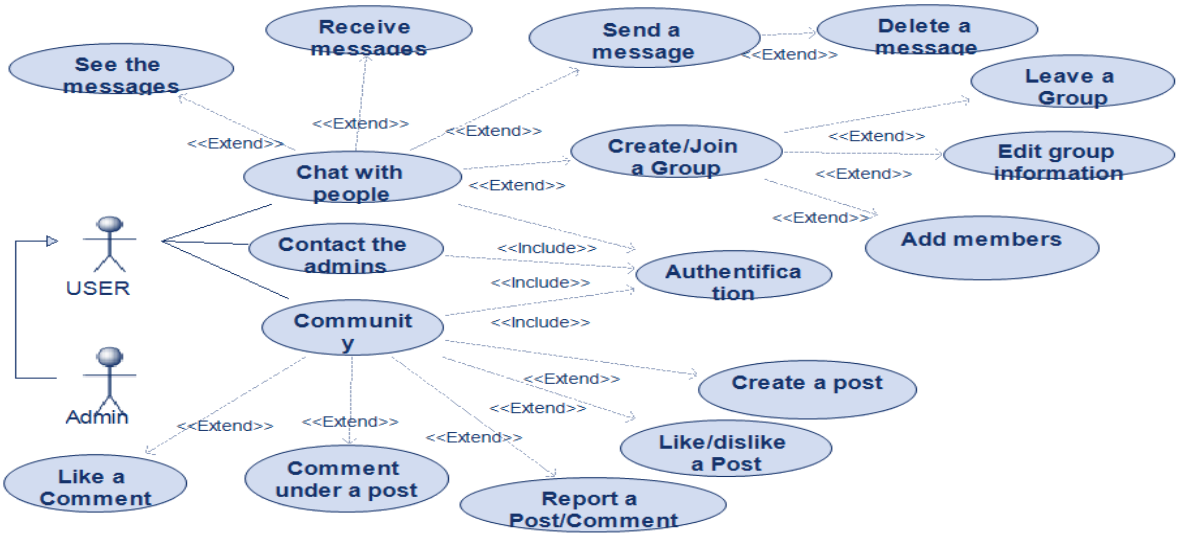


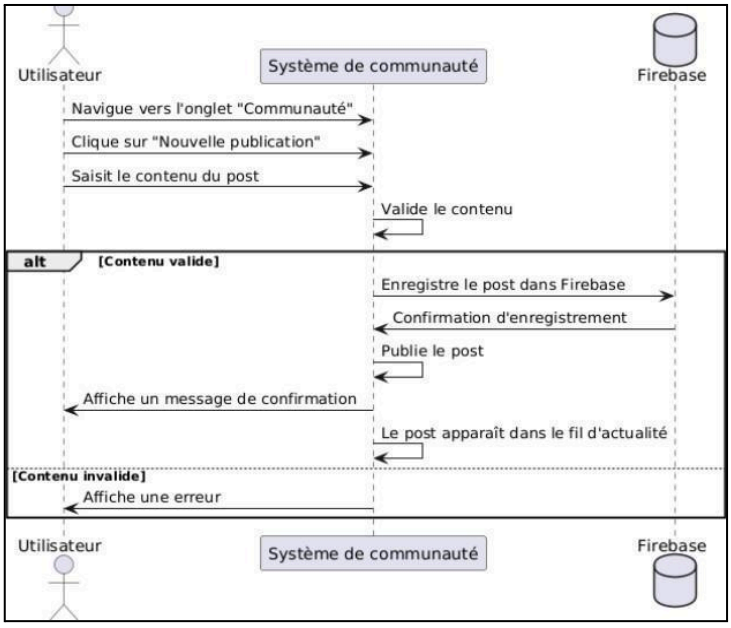
Figure 5: Community and Social Features Use cases Diagram, Admin side.

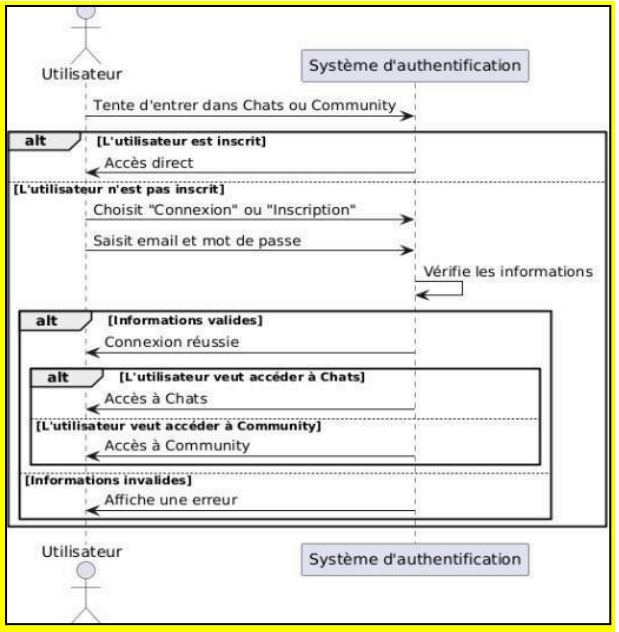
3. User Settings:

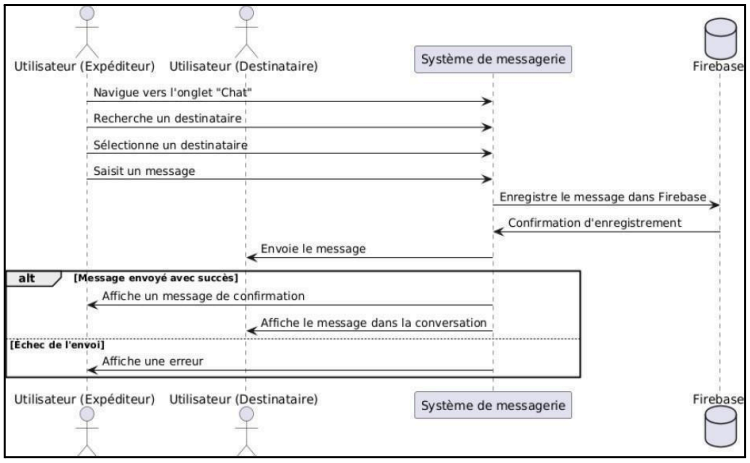
- setLanguage()/toggleTheme()/toggleNotifications(): Customize language, display theme, and notifications.
- deleteAccount()/updateAccount(): Delete or modify the user account.

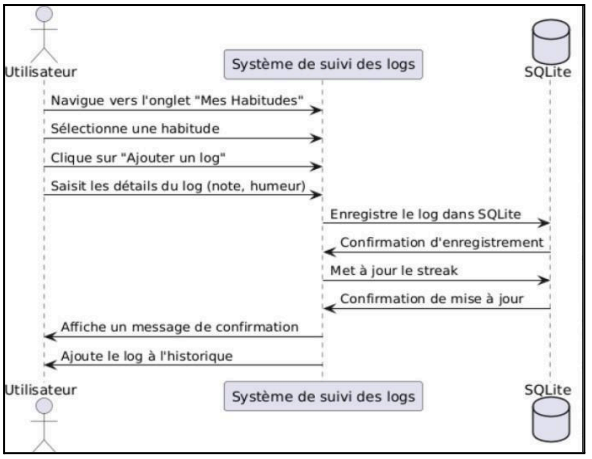
c. Sequence Diagrams

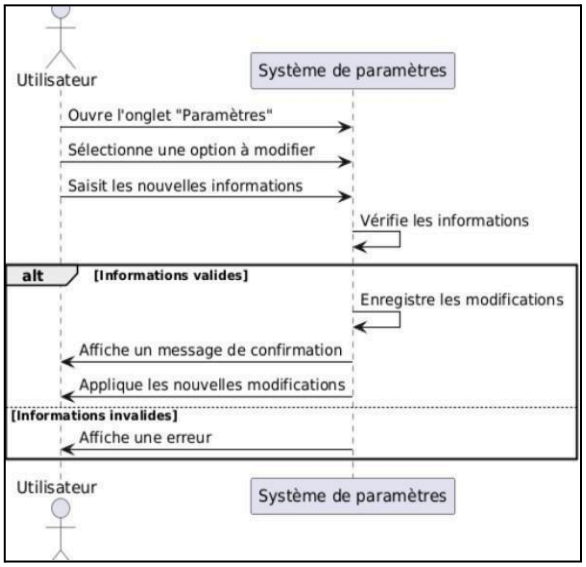
Several sequence diagrams illustrate the dynamic behavior of the application. Here are a few examples:











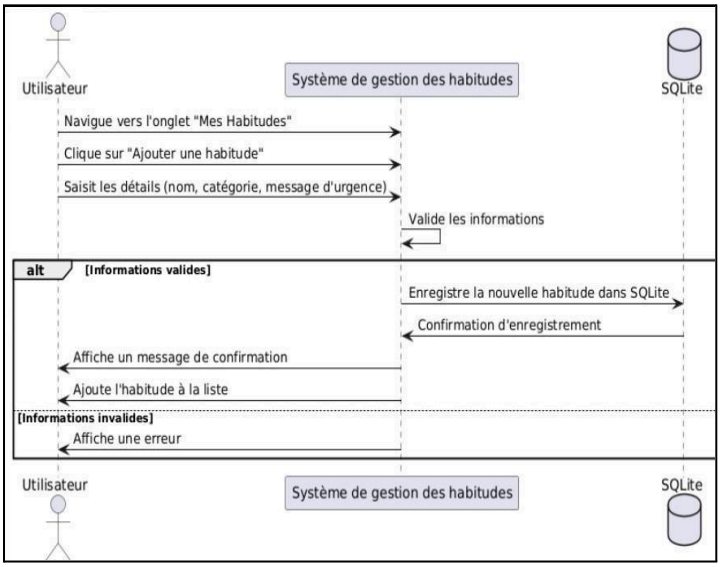


Figure 6: Principals Sequence Diagrams of ALTER application.

- **Authentication**: User logs in, credentials are verified, and access is granted.
- **Messaging**: Sending and receiving messages using Firebase.
- **Community**: Posting and retrieving posts with real-time updates.
- **Habits**: Adding a habit, tracking logs, and updating streaks using SQLite.
- **Settings**: Saving user preferences such as language, theme, or notifications.

These diagrams provide insight into how different components interact in real time to support the application's functionality.

4. Implementation

The implementation of our ALTER application was carried out in several phases, described as follows:

1. User Interface Development

Authentication Screen:

The following figure shows the authentication screen presented when a user already has an account:

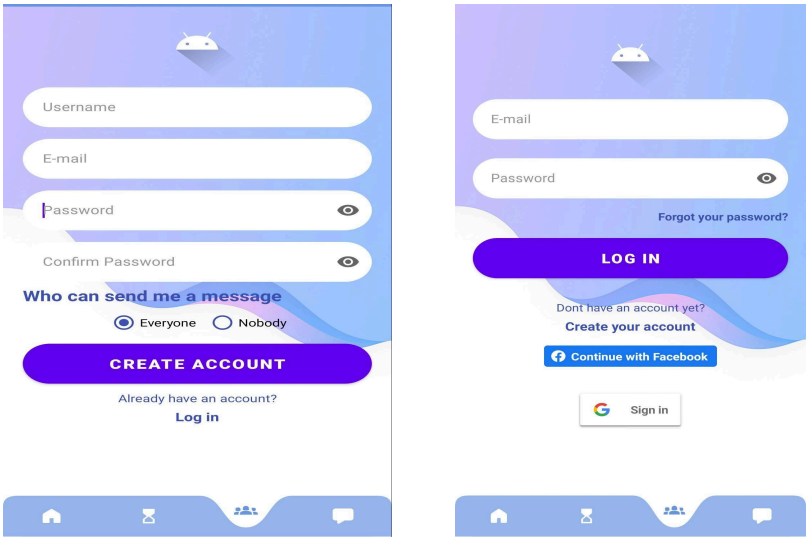


Figure 7: Login Interface for Existing Users

If the user does not have an account, they will be prompted to create one:

Home Tab:

Once authenticated, the user is redirected to the **Home** tab.
At the bottom of the screen, a **navigation bar** allows the user to switch between tabs. And right when you access to the home tab this is what displays:

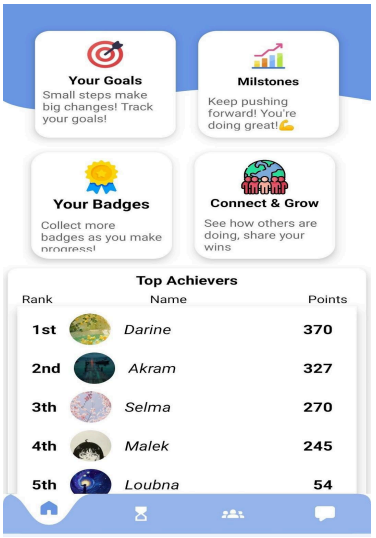


Figure 8: Home Tab.

From here, they can:

- collects **badges** by being active on the community side
- check the **progress**
- check the **leader board** of the top users
- Track their **goals** by setting:
    - A goal name,
    - A description,
    - And their current progress.

Here is an example of **your goal** interface:

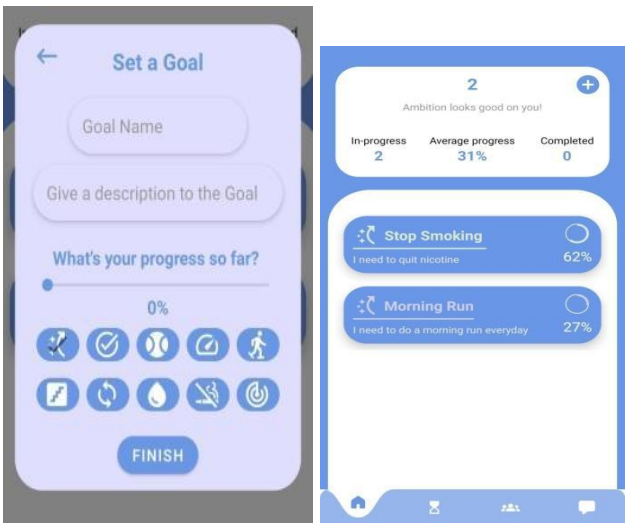


Figure 9: Goal Creation and Tracking Interface

Users can add multiple goals by clicking the **"+"** button at the top of the screen.

To monitor their progress, users can click on the **"milestones"** widget available on the home screen:

Figure 10: Milestone Tracking Widget

To check their badges and their progression as an active user, user clicks on the "**your badges"** widget on the home screen.

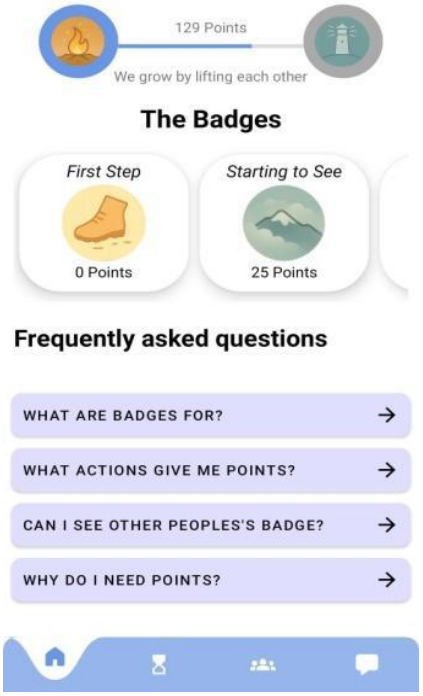


Figure 11: Badge Collection and User Progress Interface

Recovery Tab

In the **Recovery** tab, users can:

- Add their bad habits,
- Indicate how these habits affect their daily lives,
- Track how long they've resisted the habit.

- track their progress

By clicking the **“Add a Habit”** button, a form is displayed:

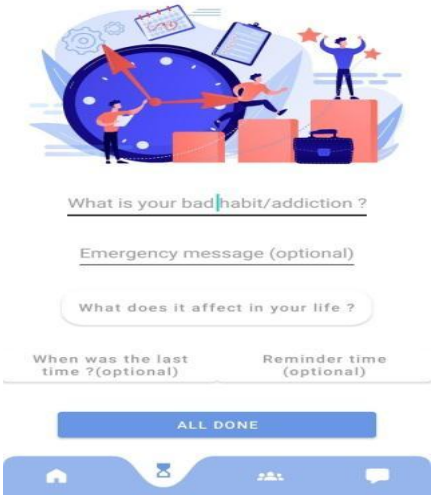


Figure 12: Bad Habit Entry Form in Recovery Tab.

Journaling System

By clicking on a bad habit (e.g., “Smoking”), the user accesses the **Journaling interface**. This feature allows:

- One log per day to update progress,
- Streak tracking (reset to zero if a day is missed),
- Logging mood and notes.

Figure 13: Journaling Interface for Habit Tracking.

Community Tab

In the **Community** tab, users can:

- Create posts anonymously or identifiably by clicking the **"+"** button and apply a category to it,
- Like or dislike posts,
- Comment on posts,
- filter the posts according to the category for a customizable experience
- Like or report comments and posts that violate platform rules.
- To filter posts, post moderation is applied to ensure a respectful and safe space.
-

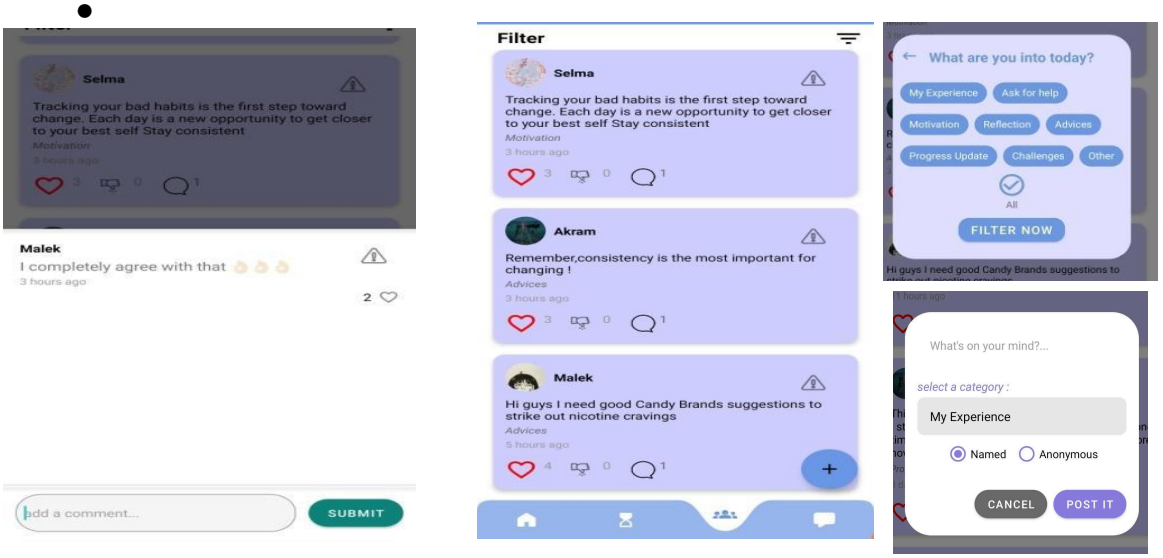


Figure 14: Community Tab with Post Creation and Interaction Options.

Chats Tab

The **Chats** section contains two interfaces:

My Chats

- Displays private messages and user-created groups and user-member-groups.
- Users can search for other users and initiate conversations via the **"+"** button.

Figure 15: Private Chat and Group Messaging Interface.

Groups

- Displays publicly or private accessible groups, each dedicated to a specific theme or addiction.
- Users can create a group by clicking the **"+"** button.

- And a filter for customizable research.

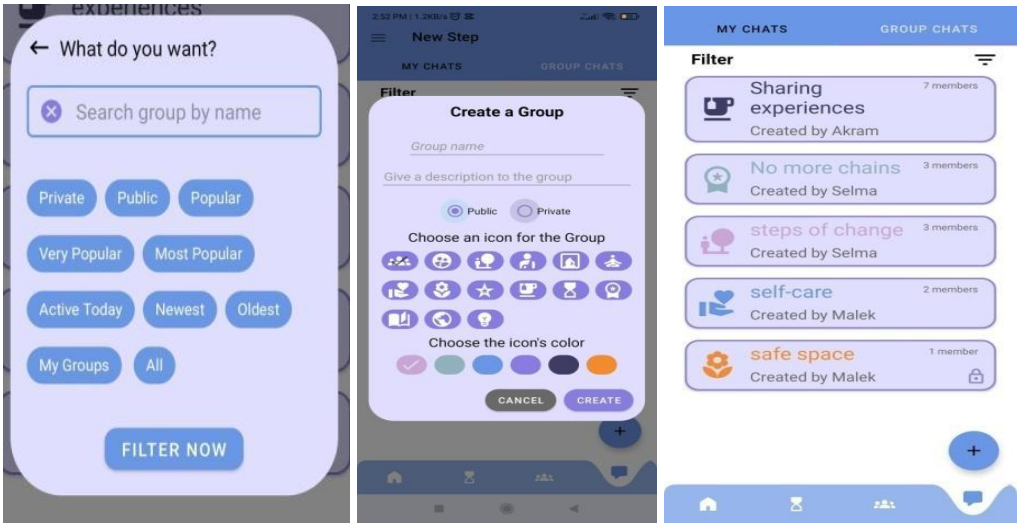


Figure 16: Public and Private Group Management Interface.

Administrator Interface

To ensure platform integrity, a **moderation system** has been implemented:

- Admin have access to a dedicated interface (a special button is visible only to them).
- Admin can:
    - View **reported accounts** and **comments**,
    - Delete posts or comments,
    - Restrict or ban user accounts,
    - Lift bans or restrictions as necessary.

    - check reports made by users concerning technical issues

Tabs include filters to facilitate moderation tasks.

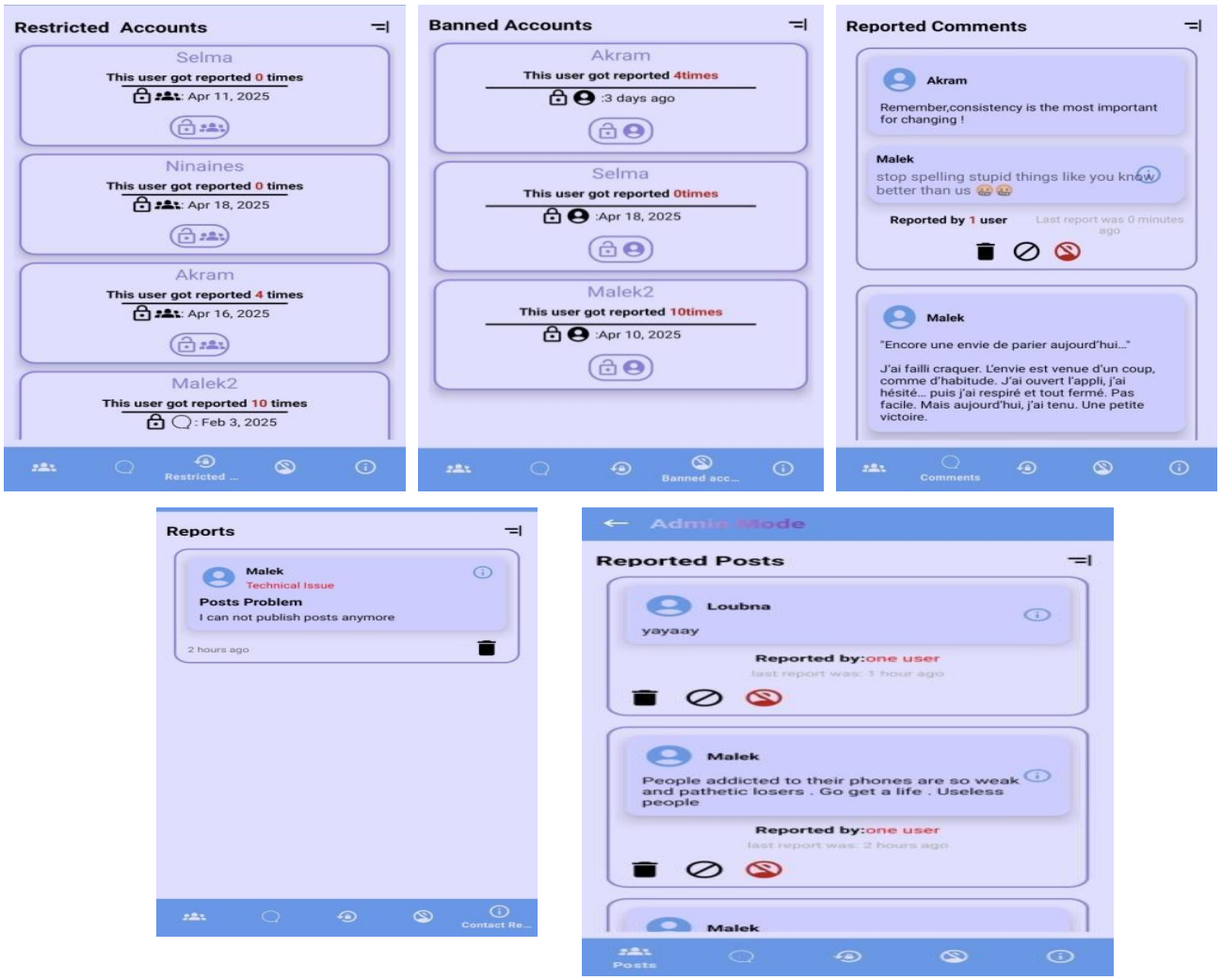


Figure 17: Administrator Dashboard for Moderation Tasks.

Reported Posts Tab

- Shows Posts that have been reported.

- Admin can delete them.

Reported Comments Tab

- Shows Comments that have been reported.

- Admin can delete them.

Restricted Accounts Tab

- Shows users who cannot post or comment.
- Admins can lift restrictions.

Banned Accounts Tab

- Displays users banned from the platform.
- Admins can unban users.

Reports Tab

It displays reports by users due to issues

User Profile

The user can access their **profile** by clicking their profile picture. The profile screen allows them to:

- Edit their profile picture, background image, and bio,
- View their posts and liked posts,
- Logout or access **Settings** via the sidebar or profile screen.

- the user can decide if anyone can send him a message or no one and can also change email and password or simply delete account , also the user can change language and the app theme

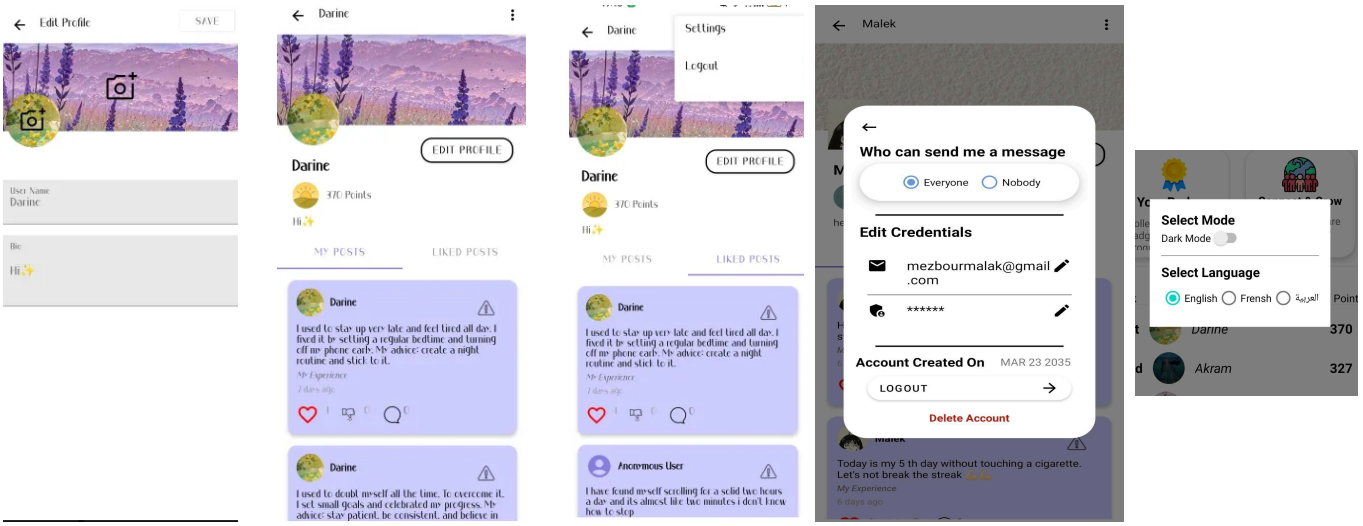


Figure 18: User Profile Management Interface.

Milo Your AI Assistant

milo is your personal AI assistant that helps the user in case of any question

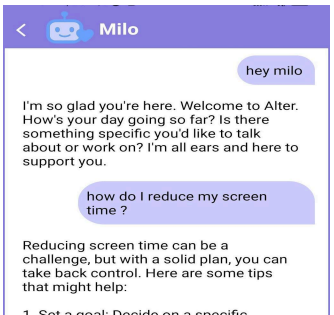


Figure 19: Milo AI Assistant .

2. Development of Main Functionalities

Key features implemented:

- Daily logging of habit progress (one entry per day),
- Posting, commenting, and reacting to content,
- Joining or creating groups,
- Real-time private messaging.

3. Database Setup

Online Database (Firebase)

Used for:

- Community posts,
- Chats,
- User profiles and authentication.

Offline Database (SQLite)

Used for:

- Local habit tracking and journaling on the device.

Key SQLite Functions:

```
public long addHabit(HabitModel habit)
public void deleteHabit(int habitId)
public List<HabitModel> getAllHabits()
public void insertDailyNote(int habitId, String date, String note,
String mood)
public boolean hasNoteForToday(int habitId, String todayDate)
public void updateDailyNote(int noteId, String newNote, String newMood)
public void updateDaysResisted(int habitId)
public List<DailyNoteModel> getNotesForHabit(int habited)
```

- set my goal in home tab

Key SQLite Functions:

```
public void addGoal(GoalModel goal);
public List<GoalModel> getAllGoals();
public void updateGoal(int goalId, String newTitle, String
newDescription, int newProgress,String newIcon);
public void deleteGoal(int goalId);
public int getTotalGoals();
public int getCompletedGoals();
public int getInProgressGoals();
public int getAverageProgress();
```

5. Conclusion and Future Improvements

While the application is fully functional, several enhancements can still be considered to improve its effectiveness and user engagement:

- **UI/UX Enhancements**: For a smoother and more aesthetically pleasing experience.
- **New Features**:
    - A space to connect with specialized therapists.
    - Improving the interface for a better user-experience

These future improvements aim to make *alter* even more responsive to user needs and more impactful in helping individuals build healthier, more balanced habits in their daily lives.